



Power notation

Task A: Notation

1) Jacob has spilt ink all over his maths notes. Identify what was under the spilt ink.

$$12^2$$

The base is 12

The exponent is 2

We say "12 squared"

It means 12×12

$$9^4$$

The base is 9

The exponent is 4

We say "9 to the power of 4"

It means
 $9 \times 9 \times 9 \times 9$

$$4^3$$

The base is 4

The exponent is 3

We say "4 cubed"

It is 64

2) Put these numbers in ascending order, smallest to largest.

$$2 + 2 + 2$$

6

$$2^3$$

$2 \times 2 \times 2$
8

$$9 \times 2$$

18

$$10 \times 4$$

40

$$9^2$$

9×9
81

$$10^4$$

$10 \times 10 \times 10 \times 10$
10 000

3) These numbers have been put in ascending order. They are missing an exponent, base or both.

$$2^{\square}, 5^{\square}, 3^{\square}, \square^2, \square^{\square}$$

a) Fill in the missing numbers using the following:

2

2

3

4

8

10

Any combination is valid.

b) Try and find a different way to complete part a)

Any combination is valid as long as it differs from part a)



Task B: Simplifying using power notation

1) Simplify the following using exponents:

a) $2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 = 2^3 \times 3^4$

b) $2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 = 2^4 \times 3^6$

c) $5 \times 2 \times 2 \times 5 \times 5 \times 5 \times 3 \times 2 \times 3 \times 3 = 2^3 \times 3^3 \times 5^4$

2) Pair up the statement with the correct simplification.

Two squared multiplied by five to the power of three.

Three squared multiplied by two to the power of five.

Two squared multiplied by two squared multiplied by seven to the power of four.

Four to the power of seven multiplied by seven to the power of four.

$4^7 \times 7^4$

$2^4 \times 7^4$

$2^5 \times 3^2$

$2^2 \times 5^3$

3) Simplify the following using **exponents** and write the expression in words:

$2 \times 2 \times 2 \times 2 \times 9 \times 9$

Two to the power of four multiplied by nine squared

$2^4 \times 9^2$

$2 \times 2 \times 3 \times 3 \times 5 \times 5$

Two squared multiplied by three squared multiplied by five squared

$2^2 \times 3^2 \times 5^2$

$2 \times 3 \times 7 \times 2 \times 3 \times 7 \times 2$

Two cubed multiplied by three squared multiplied by seven squared

$2^3 \times 3^2 \times 7^2$